Earthquake Dashboard: LLM Usage and Verification Report

Development process for a dashboard built on the public USGS Earthquake API.

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1. LLM Usage

Google Antigravity (Gemini 3.8 Flash Medium) and **OpenAI ChatGPT** were used collaboratively during development, each for distinct responsibilities:

- **Antigravity (Gemini 3.8 Flash Medium):** Served as the primary autonomous implementation assistant. Executed end-to-end fullstack development in React 19, TypeScript, and Vite; engineered the geodetic ray-casting boundary engine (`isPointInPolygon`) for India and island territories; built Leaflet mapping with OpenStreetMap tiles and magnitude-scaled markers; implemented Recharts hourly frequency and magnitude histogram pipelines; configured `AbortController` for stale query cancellation; authored the 89-test Vitest suite; and performed static code verification.
- **ChatGPT:** Assisted during initial project planning, decomposing assignment criteria into an actionable engineering checklist, structuring the 4-row filter layout strategy, and reviewing high-level UI/UX and non-predictive scientific messaging.

2. Prompting Approach

Development instructions were structured into modular, test-driven engineering milestones rather than a single ambiguous prompt:

- **Core Integration:** Query the official USGS FDSN Web Service directly using authentic GeoJSON feeds (no synthetic/mock data).
- **Telemetry Display:** Present magnitude, location place, hypocenter depth, UTC origin time, sensor status badges, and synchronized KPIs (total count, peak magnitude, average magnitude excluding nulls, shallow focus, and tsunami warnings).
- **Interactive Analytics:** Render an interactive world map with dynamic circle markers, hourly activity timelines, energy bins, and an event catalog with a synchronized slide-over telemetry drawer.
- **Geospatial Intelligence:** Resolve countries (India, USA, UK, Indonesia) with autocomplete and automatic viewport navigation, using strict geodetic boundaries to prevent false-positive inclusions.
- **Error Resilience:** Handle loading overlays, network error retries with stale-data retention, and actionable empty states.

3. Verification Against the USGS API

The dashboard was verified against live USGS telemetry on **October 19, 2026** using the official endpoint:

- **Endpoint & Query:** `https://earthquake.usgs.gov/fdsnws/event/1/query` with parameters `format=geojson`, `starttime`, `endtime`, `minmagnitude`, `maxmagnitude`, `minlatitude`, `maxlatitude`, `minlongitude`, `maxlongitude`, `limit=500`, and `orderby=time`.
- **Sampled Records:** 47 real earthquakes from a 24-hour UTC window (`M ≥ 2.5`) and live endpoint pings (`ci41349399` , M 0.72) were inspected and compared against dashboard state.
- **Field Parity:** Event ID, magnitude, location string, depth in km (handling authentic negative mountain summit depths), UTC origin timestamps, and official event URLs (`earthquake.usgs.gov/earthquakes/eventpage/{id}`) showed **0 deviation**.
- **Count Consistency:** API feature count (47) → regional spatial matches (47) → displayed count (47) → map plotted markers (47).
- **Verification Outcome:** PASS **Fully Verified (0 discrepancies).** Structure, coordinates, depth units, and event links match real USGS data.

4. Clarity and Writing Quality

Confirmed interface and codebase refinements implemented in the active project:

- **Organized 4-Row Filter Bar:** Grouped into (1) Location Search/Region/Sort, (2) Time presets (1h, 24h, 7d, 30d, custom UTC), (3) Magnitude bounds with an explicit significant toggle (`M ≥ 4.5`), and (4) Four-tier result counters.
- **Actionable Zero-Result Guidance:** When no events match active filters (e.g., UK in 24h), the map still centers on the country and presents recovery buttons (*"Expand to Past 30 Days"*, *"Lower Min Mag to 1.0+"*) instead of a broken screen.
- **Telemetry Slide-Over Drawer:** Synchronized across map markers and table rows, displaying exact coordinates, human-reviewed vs. automatic sensor badges, and one-click event ID copying.
- **Engineering Rigor:** 89 automated tests (100% passing) and 0 lint warnings (`oxlint`) ensure maintainability and scientific consistency.

5. Limitations and Corrections

- **Mid-Indian Ridge False Positives:** Naive string matching for "India" returned earthquakes from the Mid-Indian Ridge (-10° S to -45° S) in the southern Indian Ocean. Corrected via point-in-polygon ray-casting bounding mainland India, Andaman & Nicobar, and Lakshadweep.
- **Negative Hypocenter Depths:** Volcanic and mountain summit quakes (e.g., Mauna Kea) have negative depths above sea level. Initially risked rejection as corrupted data; corrected to preserve negative depths and classify them as shallow focus (<70 km).
- **Zero-Event Map Viewports:** Fixed map staying zoomed out when a selected country had zero active events by auto-navigating to the country's geodetic bounding box.
- **Non-Predictive Science:** Prominently disclosed across header, filters, and cards that the dashboard monitors **retrospective recorded observations** and does **not predict future earthquakes**.

> **Conclusion:** AI tools accelerated architecture, boundary mathematics, and testing, but rigorous manual verification against authoritative USGS records was vital to eliminate synthetic data, enforce exact geodetic boundaries, and ensure complete scientific integrity.